

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# 1.Load Dataset
```

```
# Load dataset
df = pd.read_csv("student-mat.csv", sep=';')
```

```
# 2.Explore and Clean Data
```

```
# Display first 5 rows
df.head()
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	

5 rows × 33 columns

```
# Basic Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   school      395 non-null    object
1   sex          395 non-null    object
2   age          395 non-null    int64
3   address      395 non-null    object
4   famsize      395 non-null    object
5   Pstatus      395 non-null    object
6   Medu         395 non-null    int64
7   Fedu         395 non-null    int64
8   Mjob         395 non-null    object
9   Fjob         395 non-null    object
10  reason       395 non-null    object
11  guardian     395 non-null    object
12  traveltime   395 non-null    int64
13  studytime    395 non-null    int64
14  failures     395 non-null    int64
15  schoolsup     395 non-null    object
16  famsup       395 non-null    object
17  paid         395 non-null    object
18  activities   395 non-null    object
19  nursery      395 non-null    object
20  higher       395 non-null    object
21  internet     395 non-null    object
22  romantic     395 non-null    object
23  famrel       395 non-null    int64
24  freetime     395 non-null    int64
25  goout        395 non-null    int64
26  Dalc         395 non-null    int64
27  Walc         395 non-null    int64
28  health       395 non-null    int64
29  absences     395 non-null    int64
30  G1           395 non-null    int64
31  G2           395 non-null    int64
32  G3           395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

```
#Statistical Summary
df.describe()
```

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.00
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443	3.108861	1.481013	2.29
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862	1.113278	0.890741	1.28
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.00
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000	2.000000	1.000000	1.00
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000	3.000000	1.000000	2.00
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000	4.000000	2.000000	3.00
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000	5.000000	5.000000	5.00

```
# Checking Missing Values
print(df.isnull().sum())
```

```
school      0
sex          0
age          0
address      0
famsize      0
Pstatus     0
Medu         0
Fedu         0
Mjob         0
Fjob         0
reason       0
guardian     0
traveltime   0
studytime    0
failures     0
schoolsup    0
famsup       0
paid         0
activities   0
nursery      0
higher       0
internet     0
romantic     0
famrel       0
freetime     0
goout        0
Dalc         0
Walc         0
health       0
absences     0
G1           0
G2           0
G3           0
dtype: int64
```

```
# Removing Duplicates
print("Duplicate Rows:", df.duplicated().sum())

df = df.drop_duplicates()

print("Shape After Removing Duplicates:", df.shape)
```

```
Duplicate Rows: 0
Shape After Removing Duplicates: (395, 33)
```

```
# Dataset Shape
print("Dataset Shape:", df.shape)
```

```
Dataset Shape: (395, 33)
```

```
# Dataset Datatypes
df.dtypes
```

```
0
school    object
sex        object
age        int64
address    object
famsize    object
Pstatus    object
Medu        int64
Fedu        int64
Mjob        object
Fjob        object
reason      object
guardian    object
traveltime  int64
studytime  int64
failures    int64
schoolsup   object
famsup       object
paid         object
activities  object
nursery      object
higher       object
internet     object
romantic     object
famrel       int64
freetime     int64
goout        int64
Dalc         int64
Walc         int64
health       int64
absences     int64
G1           int64
G2           int64
G3           int64
```

dtype: object

3. Analysis Questions

```
# Q1. Average Final Grade (G3)
average_grade = df["G3"].mean()
print("Average Final Grade (G3):", round(average_grade, 2))
```

Average Final Grade (G3): 10.42

```
# Q2. Number of Students Scored above 15?
high_scorers = df[df["G3"] > 15]
print("Students scoring above 15:", len(high_scorers))
```

Students scoring above 15: 40

```
# Q3.Is study time correlated with performance?
correlation = df["studytime"].corr(df["G3"])
print("Correlation between Study Time and G3:", round(correlation, 3))
```

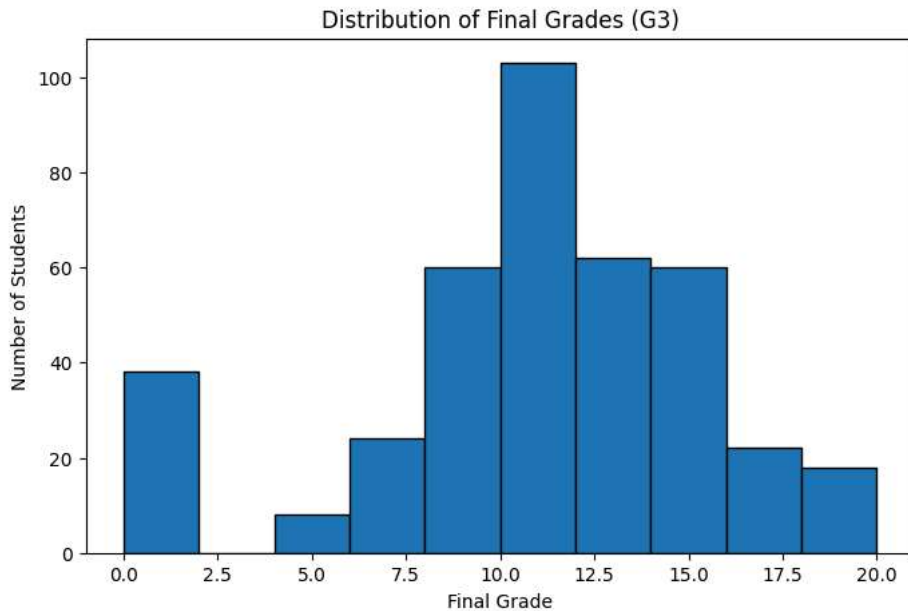
Correlation between Study Time and G3: 0.098

```
# Q4.Which gender performs better on average?
gender_performance = df.groupby("sex")["G3"].mean()
print(gender_performance)
```

```
sex
F    9.966346
M   10.914439
Name: G3, dtype: float64
```

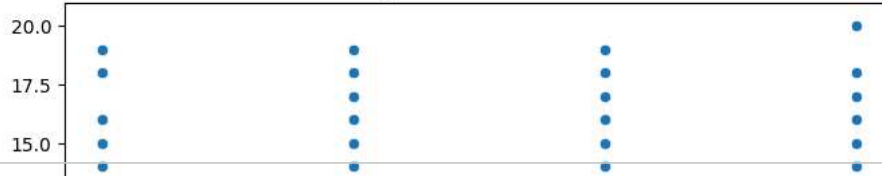
```
# 4.Visualizations
```

```
# Histogram of Grades
plt.figure(figsize=(8,5))
plt.hist(df["G3"], bins=10, edgecolor='black')
plt.title("Distribution of Final Grades (G3)")
plt.xlabel("Final Grade")
plt.ylabel("Number of Students")
plt.show()
```



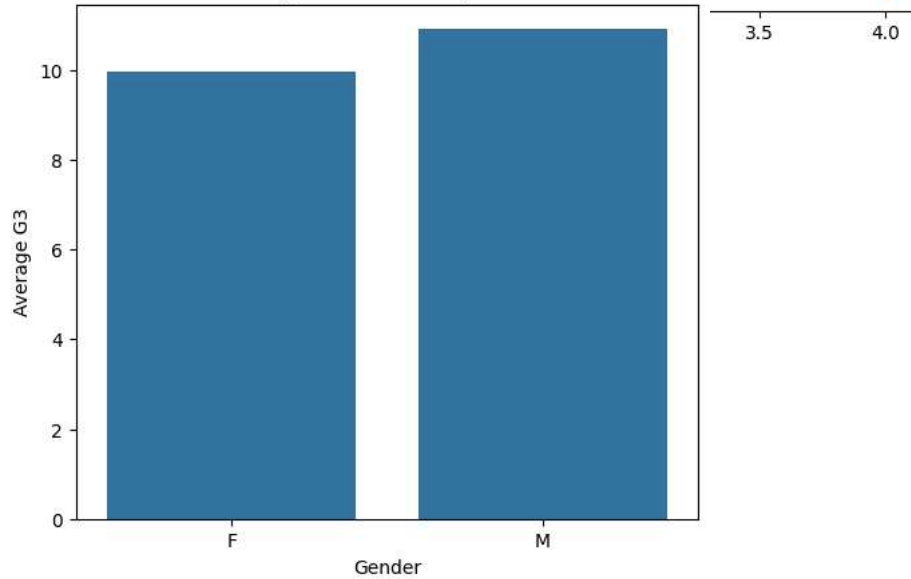
```
# Scatterplot: study time vs grades
plt.figure(figsize=(8,5))
sns.scatterplot(x="studytime", y="G3", data=df)
plt.title("Study Time vs Final Grade")
plt.xlabel("Study Time")
plt.ylabel("Final Grade (G3)")
plt.show()
```

Study Time vs Final Grade



```
# Bar chart: Male vs Female average score
plt.figure(figsize=(6,5))
sns.barplot(
    x=gender_performance.index,
    y=gender_performance.values
)
plt.title("Average Final Grade by Gender")
plt.xlabel("Gender")
plt.ylabel("Average G3")
plt.show()
```

Average Final Grade by Gender



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